TENSORING A PROPER FLAT FINITELY PRESENTED COVER WITH AN INVERSE LIMIT OF MODULES COMMUTES UP TO DQ

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof reduces to perfect test objects via the perfect generator, converts the limit to a derived limit, moves RHom through it, applies the duality and base-change identifications, and closes with the pseudo-coherent limit commutation from More on Algebra

Editorial note. The label lemma-pullback-and-limits names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the perfect generator of D_QCoh(O_X) for X proper, flat and of finite presentation over a ring (theorem-bondal-van-den-Bergh of the same chapter)
- the identity lim O_X tensor M_n = Rlim O_X tensor M_n for systems with surjective transitions (lemma-Rlim-quasi-coherent of the same chapter)
- RHom_X(E, -) is exact and commutes with products, hence with derived limits (Cohomology on Sites, global RHom section)
- the duality formula RHom_X(E, O_X tensor M) = RGamma(X, E^vee tensor^L Lf^*M) = RGamma(X, E^vee) tensor^L M for E perfect (sites-cohomology-lemma-dual-perfect-complex and lemma-cohomology-base-change of the same chapter)
- RGamma(X, E^vee) is a perfect complex of R-modules for E perfect and f proper, flat, of finite presentation (lemma-flat-proper-perfect-direct-image-general of the same chapter)
- pseudo-coherent complexes commute with tensoring and filtered limits (More on Algebra, Lemma more-algebra-lemma-pseudo-coherent-tensor-limit)

Source reference. Original reference 0A0J: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 585–585. Prerequisite pointer invoked at proof line(s) 6512; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08EU: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 4883–4883. Prerequisite pointer invoked at proof line(s) 6530; it names a standard fact of the same chapter that the counted argument

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0B91: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 6442–6442. Prerequisite pointer invoked at proof line(s) 6531; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D2L: sources/stacks-chow/tags.txt, source lines 16386–16386. Prerequisite pointer invoked at proof line(s) 6534; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08JJ: sources/stacks-chow/tags.txt, source lines 10852–10852. Prerequisite pointer invoked at proof line(s) 6523; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0B6E: sources/stacks-chow/tags.txt, source lines 14067–14067. Prerequisite pointer invoked at proof line(s) 6516; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09IS: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3891–3891. Prerequisite pointer invoked at proof line(s) 6510; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let R be a ring. Let X be an algebraic space and let $f : X \to \mathrm{Spec}(R)$ be proper, flat, and of finite presentation. Let (M_n) be an inverse system of R-modules with surjective transition maps. Then the canonical map*

$$\mathcal{O}_X \otimes_R (\lim M_n) \longrightarrow \lim \mathcal{O}_X \otimes_R M_n$$

induces an isomorphism from the source to DQ_X applied to the target.

Proof. The statement means that for any object E of $D_{QCoh}(\mathcal{O}_X)$ the induced map

$$\mathrm{Hom}(E, \mathcal{O}_X \otimes_R (\lim M_n)) \longrightarrow \mathrm{Hom}(E, \lim \mathcal{O}_X \otimes_R M_n)$$

is an isomorphism. Since $D_{QCoh}(\mathcal{O}_X)$ has a perfect generator (Theorem 09IS) it suffices to check this for perfect E. By Lemma 0A0J we have $\lim \mathcal{O}_X \otimes_R M_n = R\lim \mathcal{O}_X \otimes_R M_n$. The exact functor $R\mathrm{Hom}_X(E, -) : D_{QCoh}(\mathcal{O}_X) \to D(R)$ of Cohomology on Sites, Section 0B6E commutes with products and hence with derived limits, whence

$$R\mathrm{Hom}_X(E, \lim \mathcal{O}_X \otimes_R M_n) = R\lim R\mathrm{Hom}_X(E, \mathcal{O}_X \otimes_R M_n)$$

Let $E^\vee$ be the dual perfect complex, see Cohomology on Sites, Lemma 08JJ. We have

$$R\mathrm{Hom}_X(E, \mathcal{O}_X \otimes_R M_n) = R\Gamma(X, E^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} Lf^* M_n) = R\Gamma(X, E^\vee) \otimes^{\mathbf{L}}_R M_n$$

by Lemma 08EU. From Lemma 0B91 we see $R\Gamma(X, E^\vee)$ is a perfect complex of R-modules. In particular it is a pseudo-coherent complex and by More on Algebra, Lemma 0D2L we obtain

$$R\lim R\Gamma(X, E^\vee) \otimes^{\mathbf{L}}_R M_n = R\Gamma(X, E^\vee) \otimes^{\mathbf{L}}_R \lim M_n$$

as desired. □